

Requirement Analysis – E-Commerce Database System

1. Introduction

Flipkart is one of India's largest e-commerce platforms, connecting millions of customers with thousands of sellers across categories such as electronics, fashion, groceries, and home essentials. Behind the simple experience of browsing a product and clicking "Buy Now" lies a large, structured database system that stores and manages every piece of information involved in a transaction — who the customer is, what they are buying, who is selling it, how it will be paid for, and how it will be delivered.

This document presents the requirement analysis for the database that powers such a platform. Requirement analysis is the first and most important phase of database design: before a single table is created, the designer must understand what business processes the system supports, what data those processes generate and consume, and how different pieces of data relate to one another. Getting this phase right ensures that the resulting database is accurate, efficient, and able to grow with the business.

2. Understanding the E-Commerce Business Process

Before data can be organized into a database, it is necessary to understand the sequence of real-world activities that generate that data. For a platform like Flipkart, the business process can be broken down into the following stages:

2.1 Customer Registration and Profile Setup

A new user creates an account by providing basic identity details (name, email, phone number) and a password. The system stores this information so the customer does not need to re-enter it on every visit, and so the platform can personalize recommendations and communicate order updates.

2.2 Product Browsing and Search

Customers search or browse a catalogue of products listed by many independent sellers. Each product carries details such as price, available stock, category, and

seller information. The system must retrieve this data quickly and accurately, which is only possible if it is organized in a structured, queryable form.

2.3 Cart Management and Order Placement

When a customer decides to purchase one or more products, those items are added to a cart and eventually converted into an order. An order may contain multiple products from potentially different sellers, each with its own quantity and price at the time of purchase. This is the stage at which customer data, product data, and a new transactional record (the order) come together.

2.4 Payment Processing

Once an order is confirmed, a payment must be recorded — whether made via credit/debit card, UPI, net banking, or Cash on Delivery. The system must track whether the payment was successful, pending, or failed, since this determines whether the order proceeds to fulfilment.

2.5 Order Fulfilment, Delivery, and Returns

The seller prepares and ships the ordered products; the platform tracks the order status (packed, shipped, out for delivery, delivered) so the customer can monitor progress. If a customer is unsatisfied, a return or refund process is initiated, which again depends on accurate records of the original order and payment.

3. Data Required for Customer Management

Customer management is the foundation of any e-commerce platform, since almost every other process — ordering, payment, delivery, and support — is tied back to a specific customer record. The following categories of data must be captured and maintained:

3.1 Identity and Contact Data

- Full name of the customer
- Registered email address (used for login and communication)
- Mobile phone number (used for OTP verification and delivery updates)
- Date of birth and gender (used for personalization and age-restricted products)
- Encrypted password / authentication credentials

3.2 Address Data

- One or more shipping addresses (a customer may have a home, office, or other address)
- Billing address, which may differ from the shipping address
- City, state, and PIN code, required for delivery logistics and tax calculation

3.3 Transactional and Behavioural Data

- Order history — the list of past and current orders placed by the customer
- Saved payment methods (tokenized, not raw card numbers, for security)
- Wishlist and cart contents
- Product reviews and ratings submitted by the customer

4. Major Entities

An entity is a real-world object or concept about which data is stored, and it becomes a table in the database. Based on the business process described above, the following entities form the core of the E-Commerce Database System. Each entity's key attributes, their data types, and their purpose are listed below.

4.1 CUSTOMER

Represents a registered user of the platform who browses products and places orders.

Attribute	Data Type	Description	Key
CustomerID	INT	Uniquely identifies each customer	Primary Key
Name	VARCHAR(100)	Full name of the customer	—
Email	VARCHAR(100)	Login ID and communication channel	Unique
Phone	VARCHAR(15)	Contact number for delivery/OTP	Unique
Gender	VARCHAR(10)	Used for personalization	—
DateOfBirth	DATE	Used for age-based offers/restrictions	—

4.2 PRODUCT

Represents an item listed for sale on the platform by a seller.

Attribute	Data Type	Description	Key
ProductID	INT	Uniquely identifies each product	Primary Key
SellerID	INT	Identifies which seller lists the product	Foreign Key
ProductName	VARCHAR(150)	Name/title of the product	—
Category	VARCHAR(50)	Product category (e.g., Electronics)	—
Price	DECIMAL(10, 2)	Current selling price	—
StockQty	INT	Number of units available	—

4.3 ORDER

Represents a purchase transaction initiated by a customer; may include multiple products.

Attribute	Data Type	Description	Key
OrderID	INT	Uniquely identifies each order	Primary Key
CustomerID	INT	Identifies the customer who placed it	Foreign Key
OrderDate	DATETIME	Date and time the order was placed	—
OrderStatus	VARCHAR(20)	e.g., Placed, Shipped, Delivered, Cancelled	—
TotalAmount	DECIMAL(10, 2)	Total value of the order	—

4.4 PAYMENT

Represents the financial transaction associated with an order.

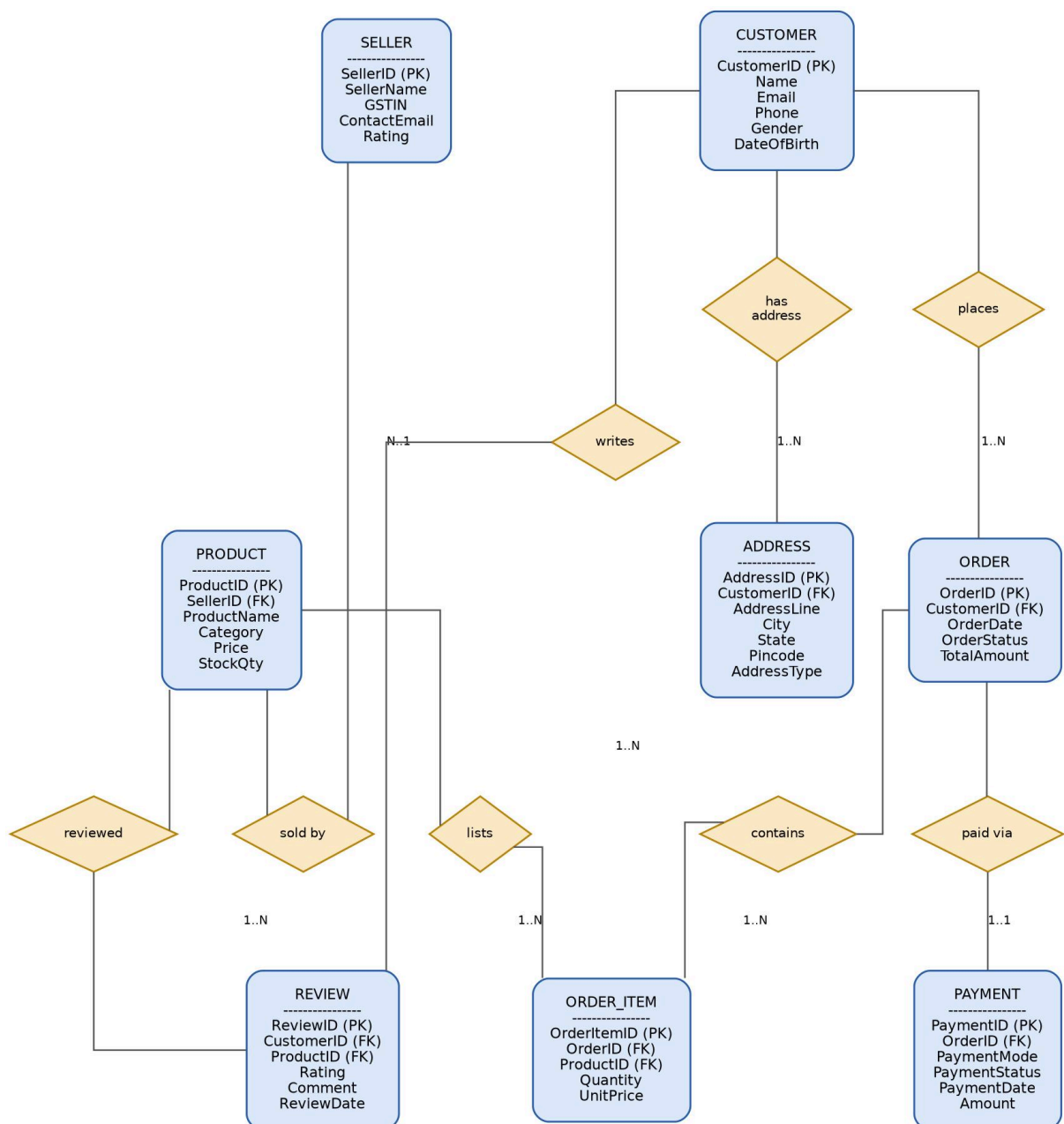
Attribute	Data Type	Description	Key
PaymentID	INT	Uniquely identifies each payment record	Primary Key
OrderID	INT	Order the payment is associated with	Foreign Key
PaymentMode	VARCHAR(20)	UPI, Card, Net Banking, COD, etc.	—
PaymentStatus	VARCHAR(20)	Success, Pending, Failed, Refunded	—
PaymentDate	DATETIME	Date and time payment was processed	—
Amount	DECIMAL(10, 2)	Amount paid	—

4.5 SELLER

Represents a merchant/vendor who lists and sells products on the platform.

Attribute	Data Type	Description	Key
SellerID	INT	Uniquely identifies each seller	Primary Key
SellerName	VARCHAR(100)	Business/store name	—
GSTIN	VARCHAR(20)	Tax registration number	Unique
ContactEmail	VARCHAR(100)	Seller's contact email	—

5. ER DIAGRAM



6. REVIEW QUESTIONS

Q1. Why does an e-commerce system require a database?

An e-commerce platform like Flipkart handles enormous volumes of interrelated, constantly changing data — millions of customers, products, orders, and payments — all of which must remain accurate and consistent at every moment. A database is required for the following reasons

- **Structured storage:** A database organizes data into related tables (Customer, Product, Order, Payment, Seller) instead of scattered files, making it easy to store, search, and update information.
- **Data consistency and integrity:** Constraints such as primary keys and foreign keys ensure that an order can never reference a customer or product that does not exist, preventing corrupted or orphaned records.
- **Fast retrieval at scale:** Indexing and query optimization allow the system to search across millions of products or orders in milliseconds — something a flat file or spreadsheet cannot do efficiently.
- **Security and access control:** Sensitive data such as passwords and payment details can be encrypted and access-restricted, which is essential for legal compliance and customer trust.

Q2. What customer details need to be stored?

- **Identity and contact details:** Name, email address, phone number, date of birth, gender, and login credentials.
- **Address details:** One or more shipping/billing addresses, including city, state, and PIN code.
- **Transactional details:** Order history, saved (tokenized) payment methods, cart and wishlist contents, and submitted reviews/ratings.
- **Account and preference details:** Loyalty/membership status, communication preferences, and account status.

These details are necessary so the platform can authenticate the customer, deliver orders to the correct address, personalize recommendations, process payments securely, and provide customer support when needed.

Q3. What customer details need to be stored?

The database must support the standard CRUD operations (Create, Read, Update, Delete) on customer data, along with additional operations specific to the business context:

- **Create (Insert):** Register a new customer account when a user signs up; add a new address when the customer saves one.
- **Read (Retrieve):** Fetch customer profile details to display on the account page; look up a customer's order history; verify login credentials during authentication.
- **Update (Modify):** Allow customers to edit their name, phone number, or address; update account status (e.g., activate/suspend); update loyalty points after a purchase.
- **Delete (Remove):** Remove a saved address the customer no longer needs; deactivate or delete an account on customer request, subject to data-retention and legal requirements.